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Can we predict stress fields without models? Recent adventures of ddfenicsx for Data-driven Identification

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Conservation laws are fundamental to physical modelling in electromagnetism, solid mechanics, and fluid dynamics. While the associated PDEs are universal, they require constitutive models – relating flux-gradient pairs (e.g., stress-strain) – to close the system. However, such models are not unique and often introduce assumptions that may bias predictions. Data-Driven Computational Mechanics (DDCM) [1] offers a paradigm shift: instead of assuming a constitutive law, it directly integrates discrete material data into the solution process. By reformulating the classical PDE problem as a mixed-integer minimisation, DDCM identifies mechanical states that satisfy equilibrium and kinematic compatibility while remaining closest to a dataset of stress-strain pairs. By operating directly on raw data, DDCM provides an alternative modeling route to both classical constitutive modeling and ML-based surrogates [2]. One interesting application of this approach lies in the intersection of experimental and computational mechanics, allowing stress field identification without explicitly postulating a parametric constitutive model, the so-called Data-Driven Identification (DDI) method [3]. This presentation shows recent advances of ddfenicsx [4], a library embedding DDCM within the FEniCSx ecosystem, preserving a familiar UFL-based workflow. Previous applications include coupled computational homogenisation [5], conservative mixed formulations for Darcy flow [6], and finite-strain problems [2], showcasing its versatility. In this talk, we focus on a new application of ddfenicsx to DDI using full-field displacement data, along with recent framework improvements: I) migration from legacy FEniCS to FEniCSx (0.10) ; ii) non-intrusive operation mode; iii) algorithmic enhancements such as the Accelerated Projection-to-Equilibrium [5] and Locally Convex Reconstruction [7] for noisy datasets and v) on-the-fly database generation [5].

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